

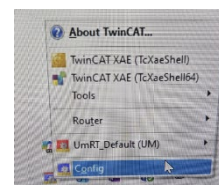
Exercise 3 Drilling machine manual auto mode with focus on good programming structure in FBD and Code according to safety rules

Equipment

TwinCAT 3, installed on a Windows computer

Preparation

- Download the project "Twincat project drill" from Canvas to your computer
- Click on the file "TwinCAT Project Drill.sln" to launch the project
- Run and select <UmRT_Default>
- Click on "Activate Configuration"  and then answer OK to download the configuration to your runtime PLC.
- If problems, look for compiling errors and set the local runtime in Config mode.



Input Signals to PLC from Process

Variable	Data Type	Comment
qxGreenLamp	BOOL	Green Lamp denoting that process have started
qxAout	BOOL	Output to activate Slider A to Move out
qxAin	BOOL	Output to activate Slider A to Move in
qxBout	BOOL	Output to activate Slider B to Move out
qxBin	BOOL	Output to activate Slider B to Move in
qxDrillDown	BOOL	Output to activate Drill to Move Down
qxDrillUp	BOOL	Output to activate Drill to Move Up
qxLampEmStopOk	BOOL	Lamp denoting all estop conditions are cleared
qxLampSafetyOk	BOOL	Lamp denoting safety sensors are working
qxLampHomePos	BOOL	Lamp denoting the robot is in home position

Output Signal from PLC to Process

Variable	Data Type	Comment
ixStart	BOOL	Start Auto Button
ixStop	BOOL	Cycle Stop Button
ixAout	BOOL	Slider A out sensor
ixAin	BOOL	Slider A in sensor
ixBout	BOOL	Slider B out sensor
ixBin	BOOL	Slider B in sensor
ixDrillUp	BOOL	Slider B out sensor
ixDrillDown	BOOL	Slider A out sensor
ixAuto	BOOL	True:=AUTOMODE, FALSE:=MANUALMODE
ixBtn1EmStopOk	BOOL	Normally Open button for emstop 1
ixBtn2EmStopOk	BOOL	Normally Open button for emstop 2
ixSafety1StopOk	BOOL	Normally Open button for Safety Sensor 1
ixSafety2StopOk	BOOL	Normally Open button for Safety Sensor 1
ixBtnReset	BOOL	Manual button for slider
ixBtnAout	BOOL	Manual button for slider
ixBtnAin	BOOL	Manual button for slider
ixBtnBout	BOOL	Manual button for slider

ixBtnBin	BOOL	Manual button for slider
ixBtnDrillOut	BOOL	Manual button for drill
ixBtnDrillIn	BOOL	Manual button for drill
iwCounter	INT	Counter of how many orders are done
iwOrderNum	INT	Order on how many orders to be done

Programming Task

Program Logic:

Make the drill function in accordance with the lecture slides L5 Programming structures.pdf

Focus on good programming structure.

- Name new Boolean variables with the prefix mx by the principle mxGood DescriptiveName
- Assign each output variable (qx) and each (mx) in one place only.
- Only one assignment in each network
- Use self-holding functions where a latch/memory function is needed.
- Create a function block for the blink function.
- Use SFC or Structure Text for the sequence.

The functionality was demonstrated during the lecture on programming structure and machine safety.



You can ignore the reset button on the left; it is only there to help you quickly reset the simulation.

Create a **video** that describes your code, shows the drilling machine in operation, and explains its functionality, with a focus on safety. You must be visible in the video sometimes. Please, keep the video short and focused!

**Upload the zipped project and the video to the Canvas assignment:
Drilling Machines Man/Auto + Video**

